



Machine Learning Analysis of Disagreement Between Wind Lidar Observations and Numerical Models

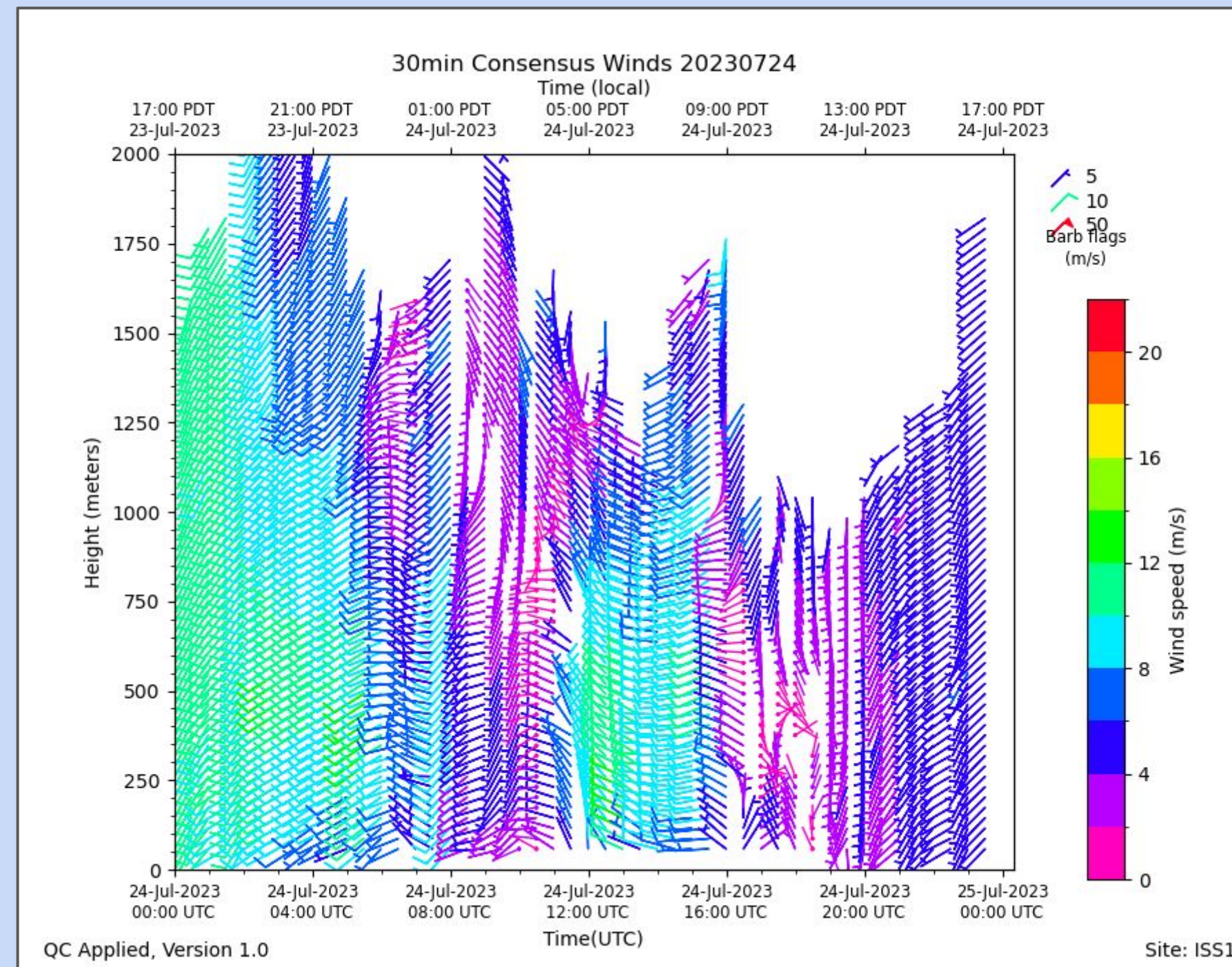
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Background & Motivation

NCAR operates a variety of instrumentation to observe atmospheric phenomena. One example of this is their **Windcube LiDAR**, which has been deployed on field campaigns such as 2023's **M2HATS** in the Nevada desert.



Above: the Windcube LiDAR itself

Left: an example VAD profile showing wind speed/direction as observed by the Windcube during M2HATS

- The **High-Resolution Rapid Refresh (HRRR)** model is a numerical model used in operational forecasting.
- ERA5** is a climate reanalysis dataset which merges model forecasts with real observations.

There exists both HRRR and ERA5 data from the M2HATS campaign in addition to the Windcube's observations.

Before proceeding, it's worth noting the differences in resolution between the models:

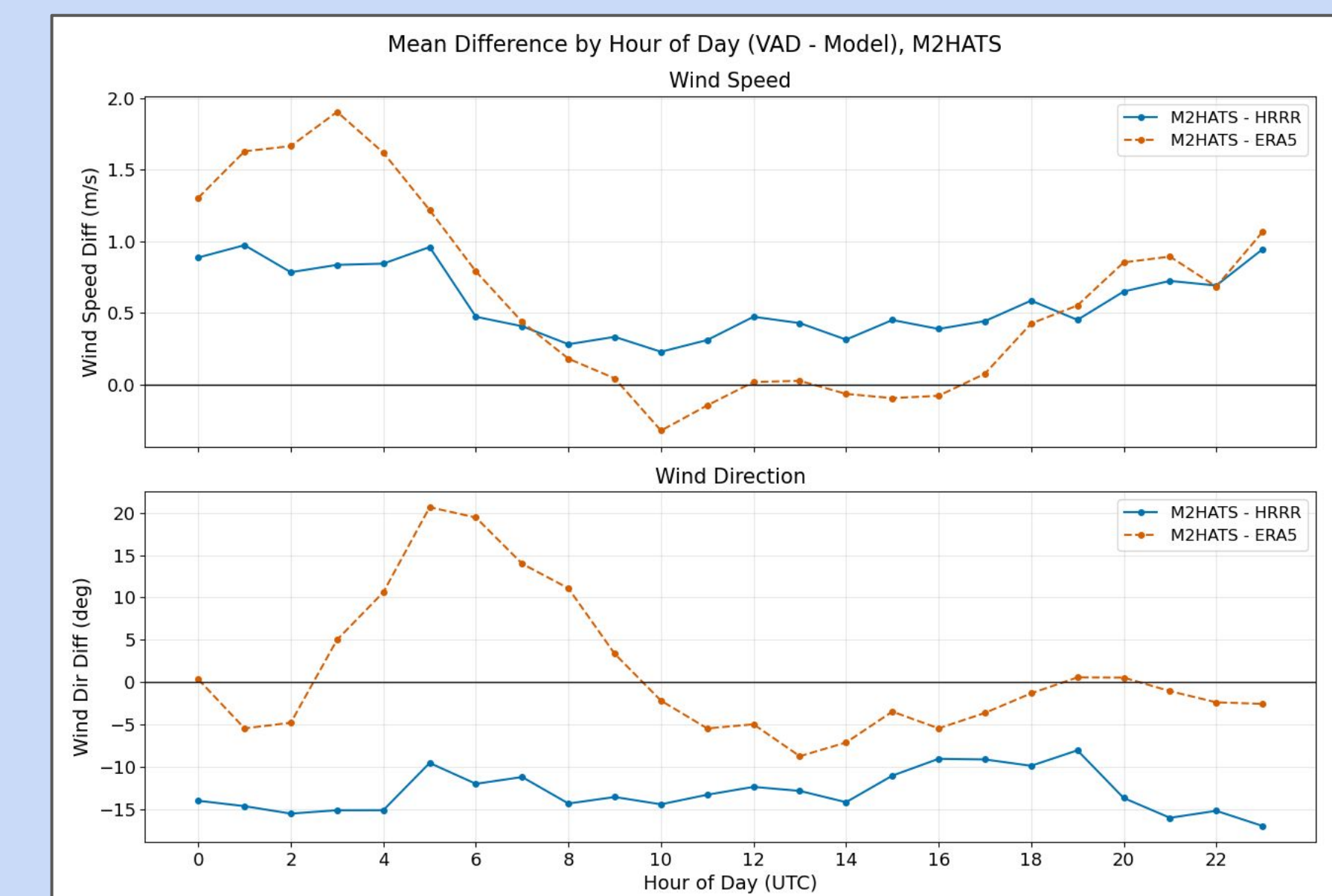
- HRRR: ~3 km (horizontal resolution)
- ERA5: ~25 km (horizontal resolution)

The Windcube has a vertical resolution of roughly 30 m.

When examining the M2HATS data, there appeared to be a relationship between the time of day and the average error between the Windcube and the HRRR/ERA5.

Thus, this project aimed to:

- See what factors influence the error between the lidar and numerical models
- Explore the helpfulness of machine learning methods in answering this question



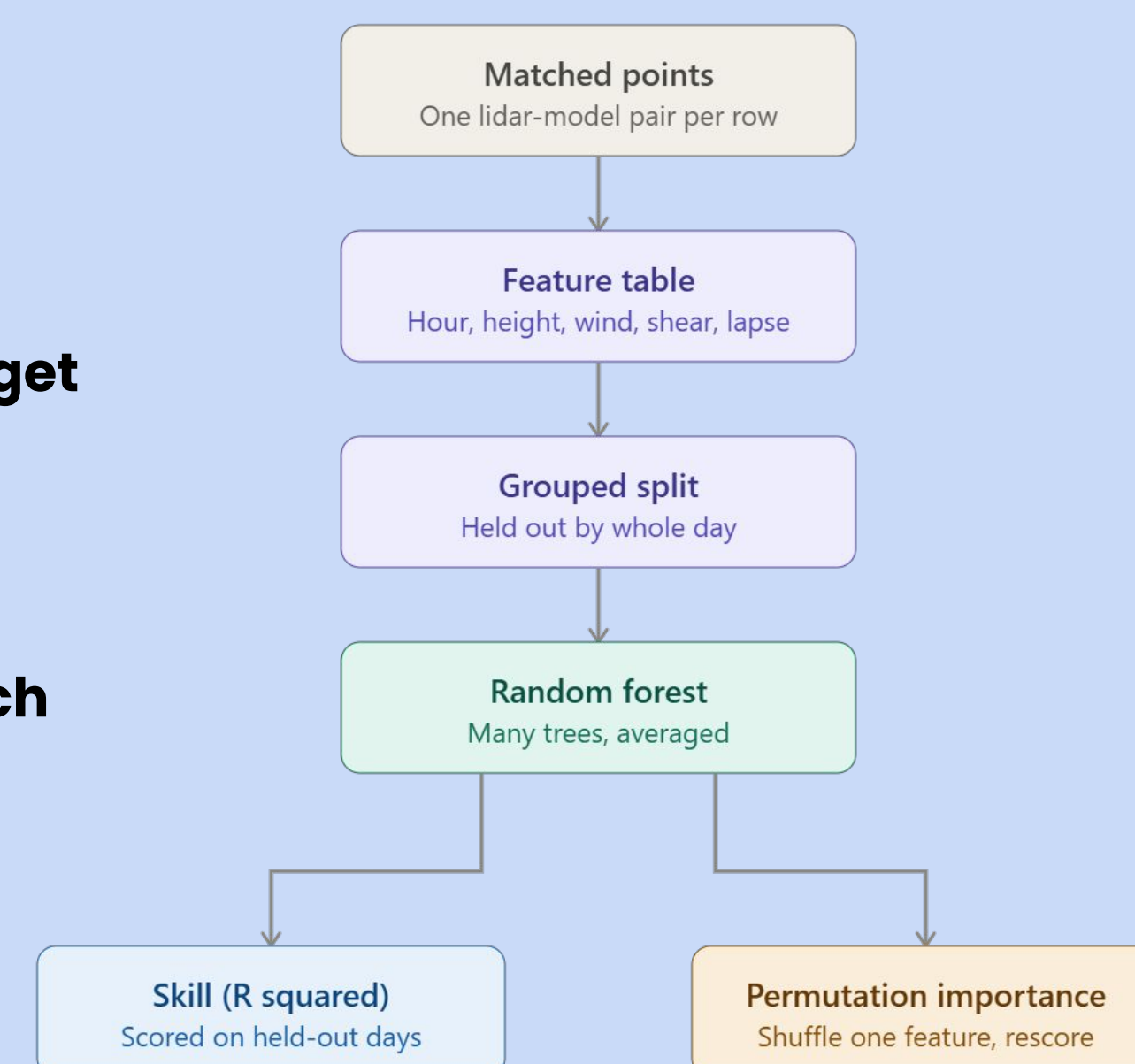
Methodology

The chosen machine learning model, **RandomForestRegressor**, is an ensemble of many decision trees. Each tree splits the data based on feature thresholds to place it into groups with similar error values, and predicts the avg. error within each final group.

The features chosen were:
hour, height, model wind speed, lapse rate, model, and shear.

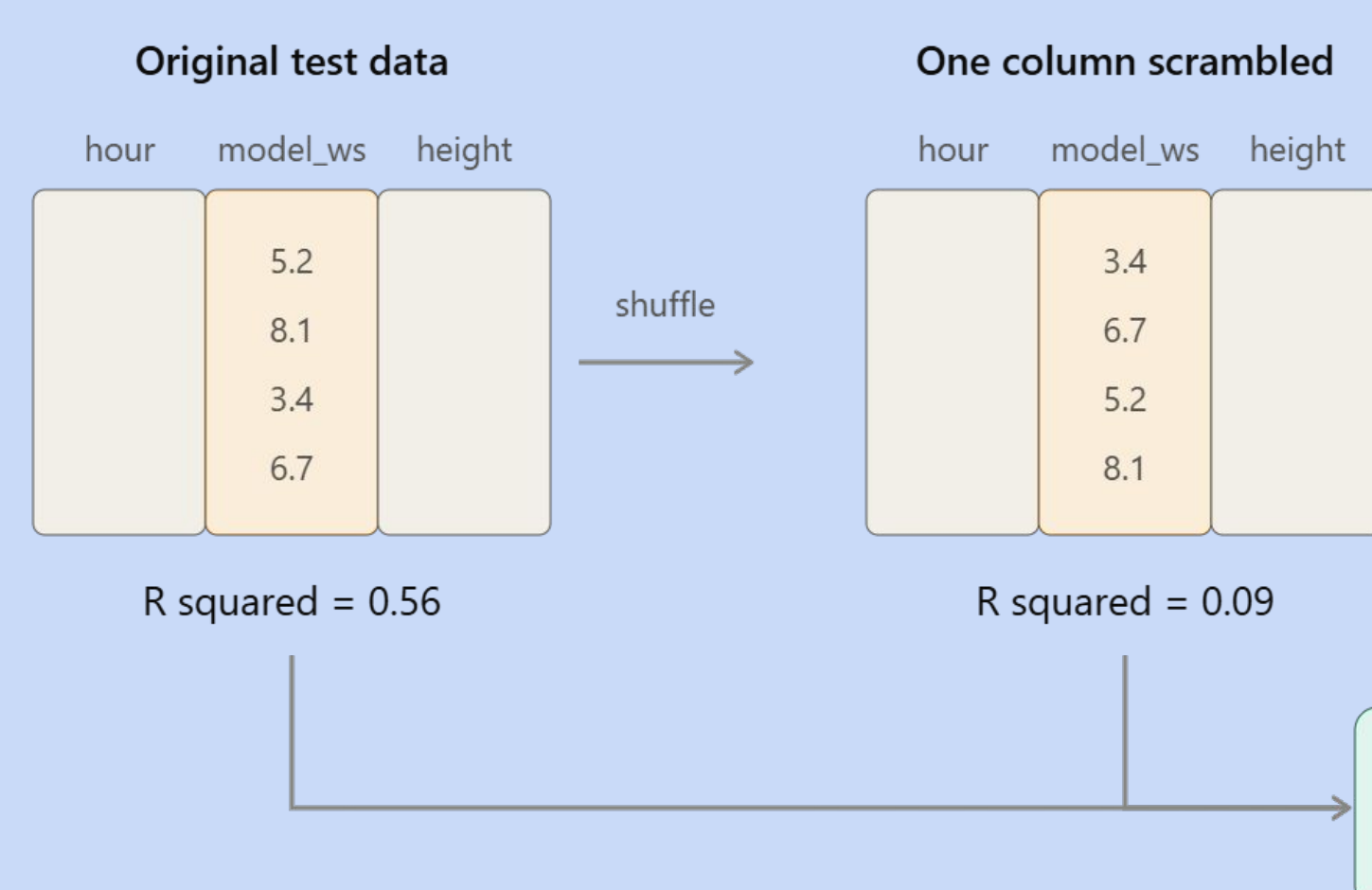
ML Pipeline

- Matched points → one row per comparison
- Feature table → predictors + target
- Grouped split → held out by day
- Random forest → many trees, averaged
- R² → skill on new days
- Permutation importance → which features mattered



Calculating Permutation Importance

- Baseline → score when using all features by R²
- Shuffle → scramble one column of features
- Re-score → measure new R² with shuffled data
- Importance → baseline score minus shuffled score
- Repeat → 10x per feature, and average out results

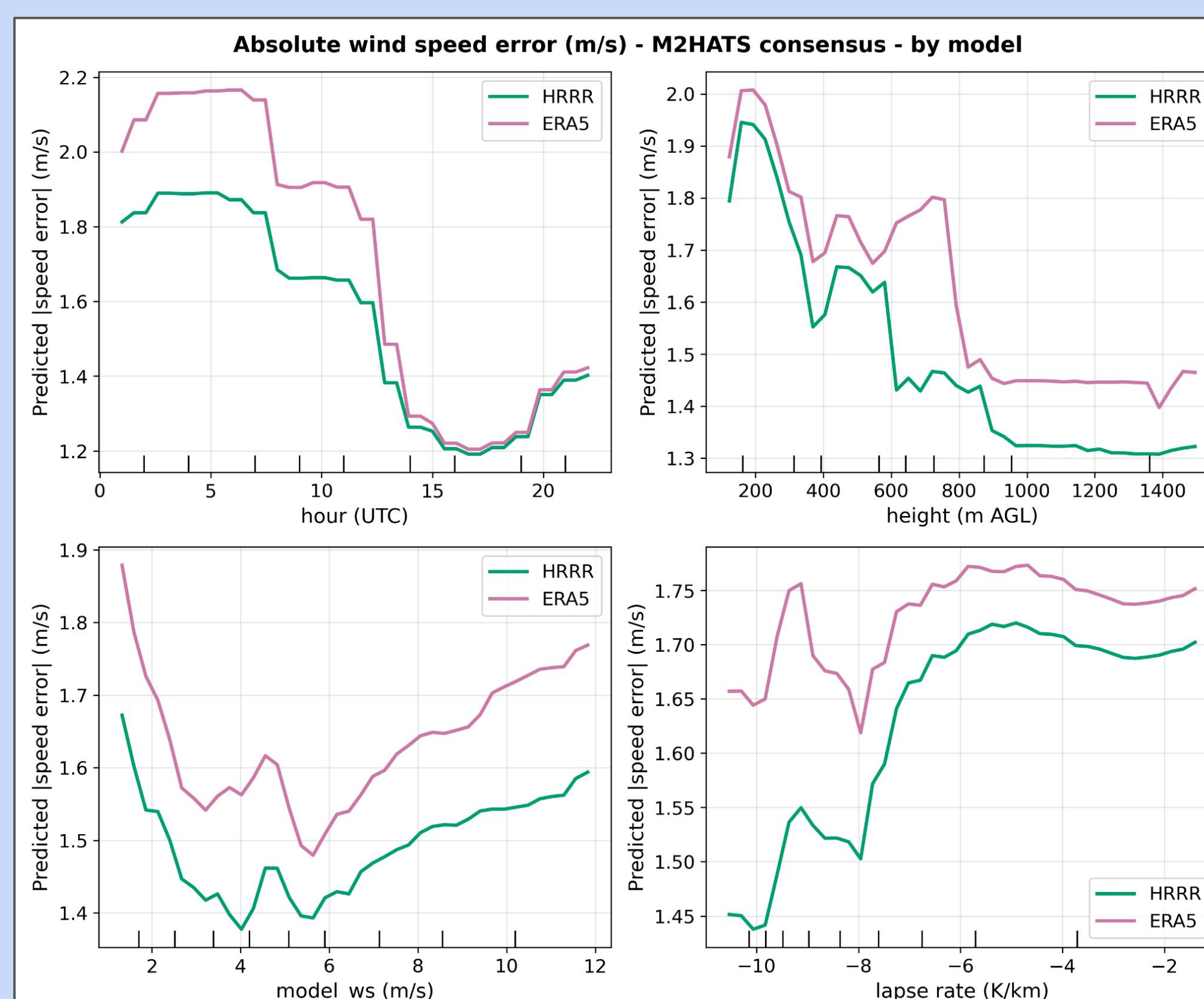


The ML model's analysis indicated that the highest permutation importance was for **hour**. Next up were **height, model_ws, and lapse rate**, with similar relative levels of importance to each other.

Feature	Permutation importance
hour	0.140 +/- 0.015
height	0.069 +/- 0.012
model_ws	0.064 +/- 0.020
lapse rate	0.055 +/- 0.013
model	0.019 +/- 0.009
shear	0.013 +/- 0.002

The overall structure of wind speed error from the model is weaker than with wind direction, but it can be gleaned that **the largest error occurs:**

- During the nighttime hours
 - Near the surface, especially <400 m AGL
 - In less unstable regimes (lapse rates above -6 K/km)
- * the pattern for model_ws is less clear



Notes:

- For both wind speed and direction, the point-to-point error remains largely unpredictable by the ML model
 - Test R² values are very low, around 0.05
- Trends in expected error shifts are robust across the grouped train/test splits and across both models.

Results

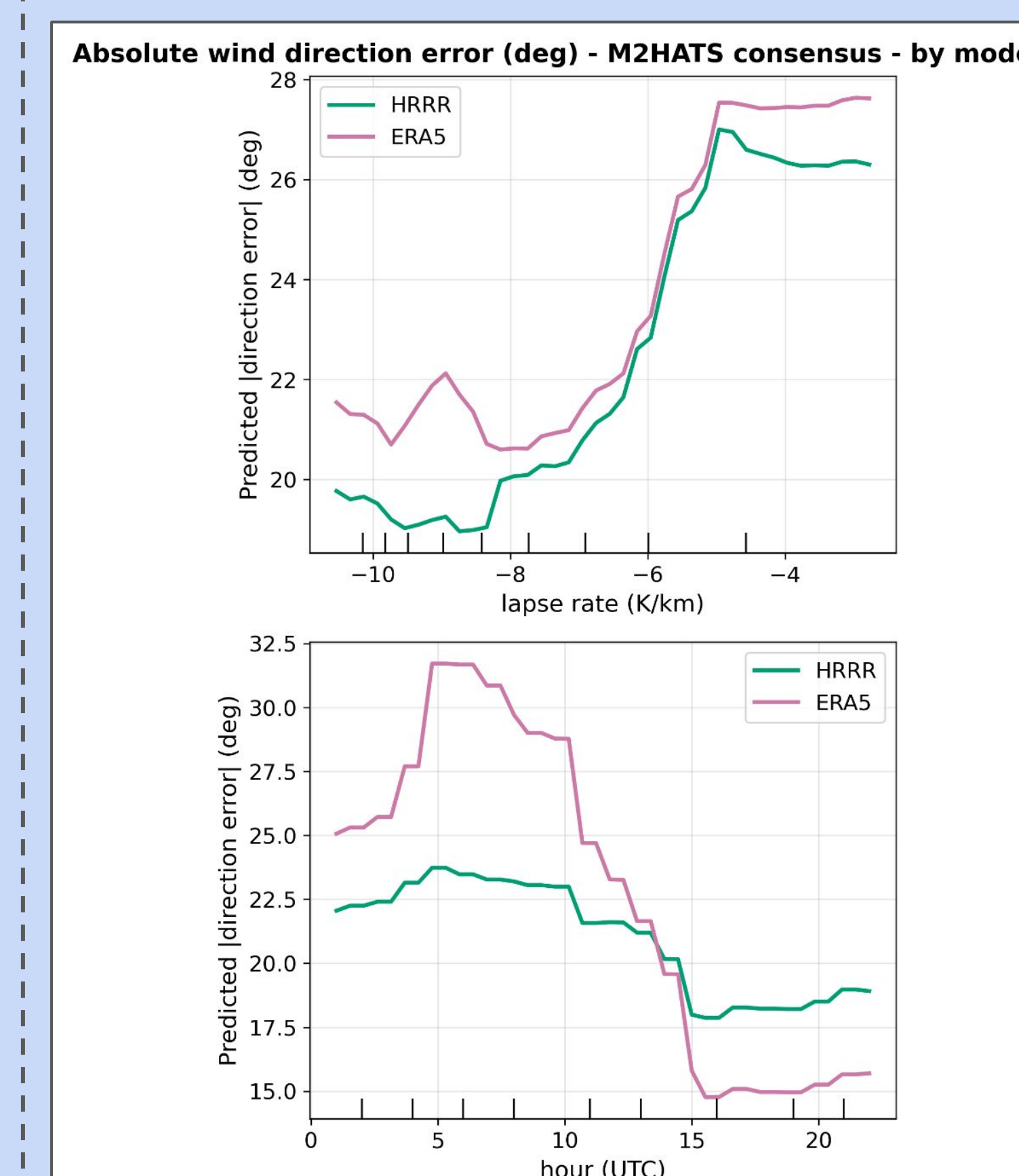
Wind Speed

Wind Direction

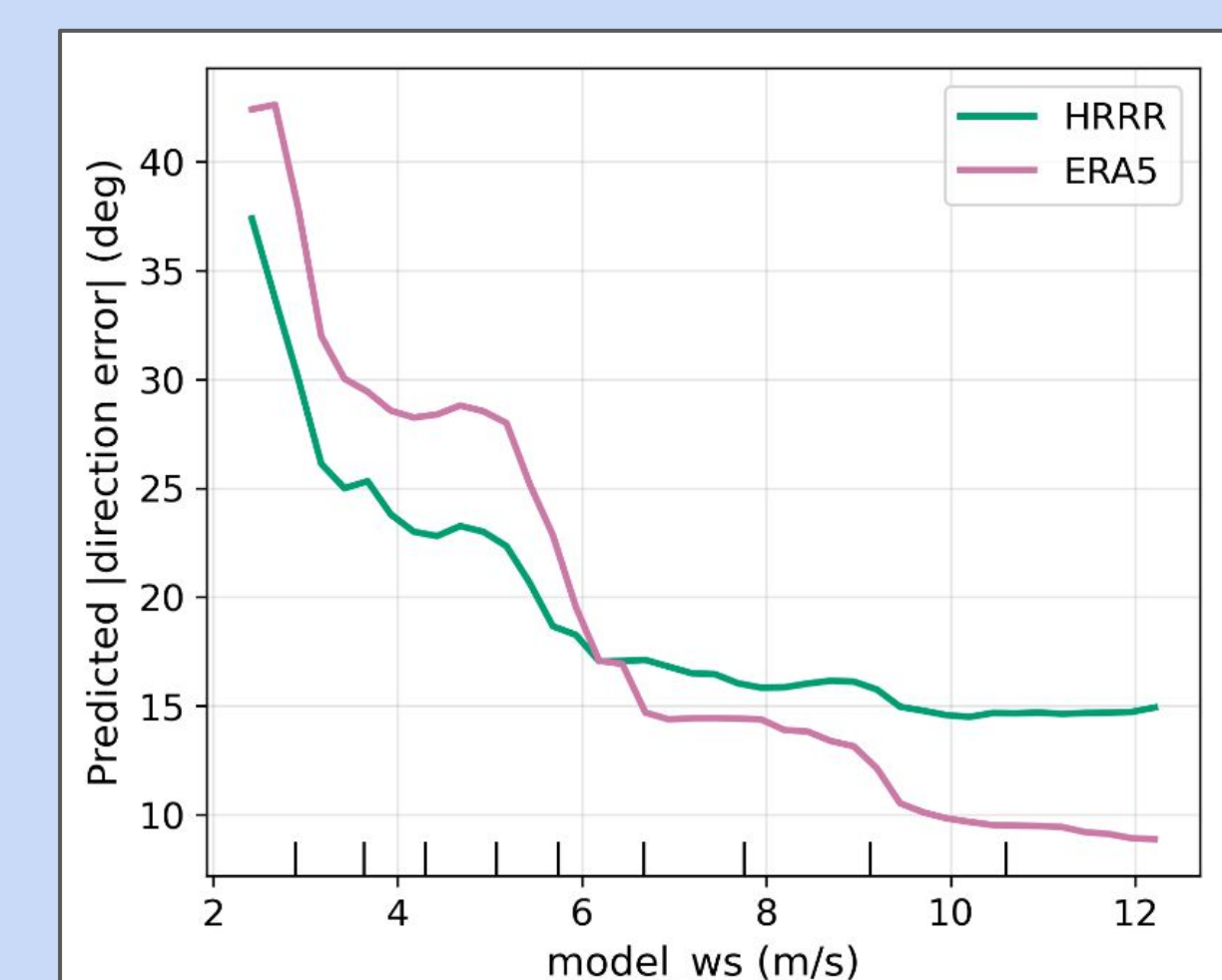
For analyzing wind direction error, the features deemed most important based on permutation importance were **lapse rate, hour, and model_ws**.

Wind direction error appears to be more strongly governed by a coherent diurnal regime, as **the largest error occurs:**

- During the nighttime hours
- With near-neutral or stable lapse rates (greater than -6 K/km)
- When winds are light (especially below 4 m/s)



Feature	Permutation importance
model_ws	0.357 +/- 0.071
hour	0.165 +/- 0.048
lapse rate	0.092 +/- 0.055
model	0.036 +/- 0.016
height	0.028 +/- 0.013
shear	0.015 +/- 0.006



Future Work

- Test these features with models other than RandomForestRegressor, try to improve R²
- Create actual filters to help with data quality control, adding onto existing filtering used based on mean carrier-to-noise ratio
- Improved quality control can be implemented in real-time for future field campaigns

Tools Used



I would like to extend a special thanks to the SUPER program for funding this work, and am very grateful for the guidance from my mentors and coordinators!

Elevator Pitch Draft:

-> [introduce name]. My project, as you may be able to gather from the title, looks at where there is disagreement between wind lidar observations and numerical models, and whether machine learning can help explain why.

On field campaigns, NCAR deploys a wind lidar, called the Windcube, which is a laser-based instrument that can measure wind speed and direction in a vertical column for a few kilometers above it. For this project, we looked at the data collected during M2HATS in the Nevada desert back in 2023. We compared those lidar winds against two models: the HRRR, which is an operational forecast model at about 3km resolution, and ERA5, a reanalysis at a coarser 25km resolution.

When beginning this project, there was the goal of exploring the helpfulness of machine learning in diagnosing error from the windcube lidar. As I plotted the average disagreement by time of day, we noticed a pattern, which was that there seemed to be greater disagreement during certain hours: in this case, generally the nighttime hours. This hinted at the disagreement being tied to something physical rather than just being random errors. Thus, we wanted to see if machine learning could help us determine what factors influence the mismatch more, and what to take away from that for future field campaigns.

-> I used a random forest, which is an ensemble of decision trees that each split the data on conditions such as height and wind speed, and together predict the expected error. Each matched lidar-model point became one row, described by six features: hour, height, modeled wind speed, lapse rate, hour, which model it was, and wind shear. The data was split by entire days, which is to say that the random forest was trained on some days and tested on separate days from that, to prevent straight up memorization of nearby points in time. Additionally, the importance of each feature was measured by permutation. Take a trained machine learning model, scramble one of the six features, and see how much worse it does. If scrambling the lapse rate, for example, has a noticeably negative effect on the predictions, then lapse rate matters a lot. If the scramble results in little-to-no change, then it didn't matter.

* One caveat or subtlety to note is that we are comparing a grid-cell average from a numerical model to a point measurement from the lidar, so some of the "error" is just a mismatch in scale.

-> As for the results, another thing I want to mention off the bat is that truthfully, the machine learning model really struggles to predict the amount of error at any single point. The R^2 was very low, around 0.05. Based on the permutation importances, however, we are led to some more interesting results. From these tables, we see that for wind speed, the most decisive features were hour, height, modeled wind speed, and lapse rate. For wind direction, lapse rate, hour, and modeled wind speed were deemed most important. With these results, the next step was to see how the expected error shifts with these conditions. This is shown through the partial dependence plots displayed, which show the machine learning model's predicted absolute error as a function of one variable, with all other features held at their observed values. This isolates each feature's individual contribution to lidar-model disagreement.

For wind speed, the signal is somewhat weak, but still consistent. The largest error occurs during the nighttime hours, near the surface, and in less unstable regimes. Modeled wind speed shows an unusual pattern where there is higher error at both lower wind speeds and higher wind speeds. The more clear results do make physical sense, however, as near the surface is where there is more turbulence and terrain effects.

For wind direction, there is a more coherent story. The largest error occurs during the nighttime hours, in near-neutral or stable lapse rates, and when winds are light. Those three things in tandem are a result of the nighttime stable boundary layer, which is a known challenge for numerical models to represent well.

One extra sanity check is that the HRRR, in almost every case, shows a smaller average error than the ERA5 (green curve sitting below the pink one). This makes sense due to its higher resolution.

-> So what's the takeaway here? Even though the machine learning model couldn't predict individual errors, it has helped to map when the largest disagreement between the wind lidar and numerical models typically occurs. Generally speaking: when it's night, winds are calm, it is stable, and near the surface. This is useful for quality control, though it should be noted that this is more relevant to understanding when the numerical models themselves are less reliable. That being said, there is overlap between the conditions for this occurring and for the lidar itself being less reliable as well. Additionally, if an algorithm using numerical model data is being used to screen lidar data in hindsight, these findings should help pinpoint when one should be more or less wary of potentially rejecting lidar data points. These practices can be implemented in future field campaigns.

-> Thanks to the SUPER program for helping fund this work, and to my mentors (credited here) as well as the program coordinators.